

Assignment 4: Vital Sign Description (FiO₂)

FiO₂ (Fraction of Inspired Oxygen) is the concentration of Oxygen in a gas mixture. In atmospheric air, FiO₂ is approximately 21% regardless of altitude. Clinically FiO₂ ranges from 21% to 100% and represents the percentage of Oxygen a patient is breathing. A patient breathing room air therefore has a FiO₂ of 21%, with higher FiO₂ values indicating the administration of supplemental oxygen.

While alone FiO₂ is not the most clinically useful parameter for triage, it is essential in assessing pulmonary gas exchange and guiding the management of hypoxemia. Hypoxemia refers to a reduction in the partial pressure of oxygen in arterial blood (PaO₂). Hypoxemia is associated with increased mortality, especially in critically ill patients, because inadequate oxygen delivery relative to tissue oxygen demand can lead to cellular damage. FiO₂ must therefore be interpreted alongside other clinical and physiologic variables, including barometric pressure and gas exchange ratio. These relationships are integrated within the alveolar gas equation (PaO₂), which helps evaluate oxygen transfer at the alveolar-capillary interface and identify causes of impaired oxygenation.